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ABSTRACT

This project is a subfield of Artificial Intelligence that focuses on Deep Learning, which involves using neural networks inspired by the human brain. Computer vision, enabled by OpenCV, has a crucial role in analysing films and individual frames. The objective is to create a resilient deep learning algorithm that can accurately identify deepfake videos, in response to the increasing worry about the improper use of artificial media. We preprocess the dataset by removing frames and specifically focusing on facial regions, utilising a collection of actual and false videos. A bespoke model, combining ResNeXt-50 for extracting features and an LSTM network for analysing temporal patterns, was trained to differentiate between authentic and counterfeit videos. The model demonstrated exceptional accuracy, efficiently detecting deepfakes with substantial levels of confidence. The results highlight the capability of advanced deep learning methods in countering digital fraud, opening up opportunities for additional research and improvement in deepfake detection technologies.